Ankit Chaurasia

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Professional Summary

Senior Software Engineer with 9+ years of experience spanning software engineering, data engineering, and AI/ML. Deep expertise in Apache Airflow, Apache Spark, Apache Kafka, Apache Flink, scalable backend systems (REST APIs, microservices, distributed systems), SQL/NoSQL databases, Snowflake, data pipelines, ETL/ELT, data modeling, data lineage, data governance, data observability, and LLMOps. Key contributor to Apache Airflow 3.0, with 130+ open-source contributions. Built engineering organizations from scratch - hiring, mentoring, and leading high-performing full-stack teams while driving technical strategy, architectural vision, and cross-functional execution. Speaker at multiple Airflow Summits; keynote at Microsoft IDC Hyderabad on advanced data engineering and LLMOps.

Skills

- **Software Engineering:** Data Structures & Algorithms, System Design, REST APIs, Microservices, CI/CD, OAuth 2.0
- **Data Engineering:** Data Pipelines, ETL/ELT, Data Modeling, Streaming Architectures, Data Lineage, Data Governance, Data Observability, SQL
- **AI & ML:** Large Language Models, LLMOps, LangChain, PyTorch
- **Languages:** Python (Proficient), C++ (intermediate), JavaScript, PHP, C, Java, Bash, LaTeX
- **Frameworks:** Apache Airflow, Apache Spark, Apache Kafka, Apache Flink, Django, FastAPI, Flask, OpenCV, OpenLineage
- **Databases & Warehouses:** PostgreSQL, MySQL, MongoDB, Weaviate, Snowflake, BigQuery, Redshift, Redis
- **Cloud & Infrastructure:** AWS, Azure, GCP, Docker, Kubernetes, AWS Glue, AWS EMR, AWS Lambda, AWS Fargate, AWS Cognito
- **DevOps & Tools:** Git, GitHub Actions, CircleCI, Swagger, Postman
- **Leadership:** Team Management, Technical Hiring, Mentoring, Architecture Reviews, Agile

Experience

- **Senior Software Engineer, Astronomer - Remote, India** Jan 2022 - Present

  Leading high-performing, delivery-focused full-stack engineering teams as a senior individual contributor - driving technical strategy, architecture vision, cross-functional execution, mentoring, and major open-source contributions to Apache Airflow.

  - Set technical strategy and spearheaded key features in Apache Airflow 3.0 - AIP-83 (directly support new use cases, such as model training and generative AI inference), AIP-72 (for multi-cloud and multi-language support), influencing cross-organizational technical direction; built a Ruff linter and config tool for seamless 2.x to 3 upgrades. These contributions were pivotal to Astronomer's $93M Series D funding.
  - Led development of Astro Runtime (Astronomer's production-grade Airflow distribution), managing 80% of releases with complex Airflow dependencies to ensure Day-0 customer support; engineered a new logging module to handle Airflow task logs via HTTP endpoints, slashing Docker image size by 80%.
  - Built Ask-Astro, an open-source chatbot for Airflow and Astronomer, using generative AI; orchestrated documentation ingestion with Airflow and vector database embedding to enable real-time retrieval with feedback loops, showcasing practical LLMOps.
  - Mentored and developed engineers across multiple teams, providing technical guidance, architecture reviews, and career development support; established organization-wide engineering best practices and standards for AI coding assistants (Cursor, GitHub Copilot) to boost developer productivity.
  - Improved engineering efficiency by instituting nightly build workflows to catch regressions early (reducing manual testing by 90%) and driving organization-wide process improvements across engineering teams.
  - Developed apache-airflow-providers-transfers, a unified interface replacing 100+ specialized data transfer operators, and co-created the Astro Python SDK to streamline ETL development by abstracting Airflow boilerplate; these frameworks are now adopted organization-wide and by the broader Airflow community.
  - Enabled deferrable mode for 80+ Airflow operators/sensors, freeing up worker slots during external waits (via lightweight async triggers) and reducing compute costs by up to 70%.
  - 130+ open-source contributions to Apache Airflow spanning new operators, providers, core features, bug fixes, and documentation improvements.

- **Senior Software Engineer, Wadhwani AI - Mumbai, India** Jan 2019 - Dec 2021

  Joined as first engineer; built and scaled the engineering organization to 25+ members, setting technical strategy, architectural vision, and infrastructure standards while leading delivery-focused teams and driving clarity with product and program teams.

  - Hired and scaled the engineering team from scratch - conducting technical interviews, establishing hiring standards, defining the technology stack, development practices, and infrastructure standards that scaled across multiple projects.

- Designed, developed and deployed a pest management system which runs a PyTorch-based model on the phone/cloud to identify the harmful pests, alert farmers about infestation and reduce crop loss in cotton farming. Currently being used by over 11,000 farmers.
- Part of the only Indian team to win a $2 million grant in the Google AI Impact Challenge, which recognized just 20 teams globally and was featured at Google I/O and Google for India.
- Developed pipeline for data collection ODK, annotation LabelMe, model deployment (online on-cloud and offline on-device), analytics dashboard along with open APIs for third-party integration and Django backend.
- Built annotation studio platform orchestrating annotation jobs across multiple vendors using Apache Airflow, FastAPI, and React, streamlining data management and scheduling workflows.
- Established technical architecture and developed reusable micro-services (authentication, geospatial directory, emailing) using Django, AWS, and OAuth 2.0, serving multiple teams across the organization and eliminating duplicate infrastructure; these services became foundational components for all engineering teams.
- Reduced video buffering time by 75% on open-source CVAT annotation tool by implementing Redis caching during data ingestion.
- Built COVID-19 mobility dashboard integrating COVID-19 Mobility Data Network and Facebook Data For Good, enabling 11 state governments in India to optimize emergency protocols based on epidemic spread patterns.

- **Software Development Engineer, DreamVu - Hyderabad, India** Dec 2017 – Jan 2019
  - Developed end-to-end modular video acquisition and processing software for VR/AR and machine vision applications (in-house camera) along with SDK for third-party integration, helping secure $0.5M angel investment from SRI Capital.
- **Software Engineer, AapkaPainter - Bangalore, India** Aug 2016 – Nov 2017
  - Built CRM, task management, and sales automation systems; integrated Razorpay payment gateway processing 10M+ INR monthly; developed Flask API for automated floor plan measurement extraction using computer vision.

Open Source Contributions

- apache-airflow, ask-astro, ruff, apache-airflow-provider-transfers, astro-sdk, astronomer-providers

Talks and Publications

- Beyond Execution Dates: Empowering inference execution and hyper-parameter tuning with Airflow 3 | Apache Airflow Summit 2025 Seattle
- Seamless Airflow Upgrades: Migrating from 2.x to 3 | Apache Airflow Summit 2025 Seattle
- Mastering Advanced Dataset Scheduling in Apache Airflow | Apache Airflow Summit 2024 San Francisco
- Data pipelines from ETL to ML to LLM: Powered by Apache Airflow | Microsoft IDC Hyderabad 2023
- Love for Writing Deferrable Operators — Why and How to Defer? | Apache Airflow Summit 2022
- Pest management in cotton farms: an AI-system case study from the global South | KDD 2020 [Research Paper]
- A real-time travel planning algorithm via multimodal transport network for ATIS | B.Tech thesis 2016 [Thesis]

Education

- **B.Tech. Computer Science and Engineering** 2012–2016
  National Institute of Technology Warangal, Warangal, India
- **Technology Entrepreneurship Programme** 2014–2016
  Indian School of Business, Hyderabad, Hyderabad, India

Achievements

- **Winner, Design Challenge, ISB Hyderabad** (presentation) – Founded award-winning startup building real-time position reporting system for buses without GPS; pitched to Telangana IT minister and received seed funding offer from TSRTC.

Extracurricular and Leadership

- **Coordinator, Quality Control & Management, Technozion 15** (Aug–Jan 2016) – Secured ISO 9001:2008 certification and UNESCO patronage for South India's second largest technical fest; generated 1.5Cr INR ($200k) revenue; managed 150+ events across coordination, quality control, and risk management.
- **Teach for India** – 70+ hours of teaching community service in local underrepresented schools.

Last updated: February 6, 2026